

Cooper Wisener

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EDUCATION

University of Wisconsin–Madison

Madison, WI | Expected May 2027

B.S. Data Science, Minor in Business — GPA: 3.84/4.0

Dean's List: Fall 2023, Spring 2024, Fall 2024, Spring 2025, Fall 2025, Spring 2026

TECHNICAL SKILLS

Languages: Python, R, Java, HTML

Machine Learning & Stats: XGBoost, Random Forests, glmnet, NLP, Markov Chains, Monte Carlo Methods

Tools & Libraries: pandas, ggplot2, Flask, Git, Apache, Stata, Excel

PROFESSIONAL EXPERIENCE

Data Science Research Assistant | *Niedenthal Emotions Lab, UW–Madison* Sept 2025 – Present

- Built automated data pipelines in R and Python to clean, organize, and aggregate raw experimental data from psychological studies, reducing manual processing time and improving reproducibility.
- Applied NLP and supervised machine learning to quantify sentiment in subject responses, mapping text to valence and arousal dimensions.
- Designed and administered behavioral studies end-to-end, from IRB-compliant protocol design to data collection, supporting multiple research projects.

Farm Hand & Greeter | *Peck and Bushel Orchard* Aug 2022 – Present

- Maintained a structured, process-driven work ethic, managing equipment upkeep and coordinating irrigation, spraying, and pruning schedules across seasonal operations.

PROJECTS

Attack or Contain | *GitHub* Oct – Dec 2025

- Applied change-point detection to classify NFL defensive player reactions to pass plays in zone coverage, identifying behavioral thresholds in tracking data.
- Trained an XGBoost multiclass classifier to predict whether a defender should attack the ball or contain the receiver.

Pass Catcher CPOE Model | *GitHub* Sept – Oct 2025

- Built a Random Forest model using the ranger package to predict expected completion percentage (xCP) for NFL pass catchers, controlling for route type, defender positioning, and game situation.
- Derived completion percentage over expected (CPOE) to identify receivers consistently outperforming or underperforming their statistical baseline.

4th Down Decision Simulator — Markov Chains | *GitHub* Oct – Nov 2025

- Constructed a Markov Chain model of NFL in-game state transitions to estimate shifts in win probability from 4th-down decisions.
- Used simulation results to benchmark Matt LaFleur's 4th down tendencies against league-wide averages, providing data-driven insight into coaching aggressiveness.

DelphiAI — Spotify Recommendation App | *GitHub* May – Sept 2024

- Developed a full-stack Flask web application integrating the Spotify OAuth API to surface personalized listening history, top artists, and track statistics.
- Incorporated the ChatGPT API to generate music recommendations based on user search queries and historical listening behavior.

RELEVANT COURSEWORK

STAT 340: Data Science Modeling II • STAT 479: Sports Analytics • CS 540: Intro to AI • OTM 300: Operations and Supply Chain Management • Math 340: Linear Algebra • CS 320: Data Science Programming II